

THE SINGULARITY CATEGORY OF A (COMMUTATIVE) GORENSTEIN RING

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ABSTRACT

Given a commutative Noetherian ring R , Ragnar-Olaf Buchweitz introduced the notion of the singularity category of R (under a different name). Buchweitz showed that when R is Gorenstein, the singularity category can be better understood through maximal Cohen-Macaulay modules and complete resolutions. We will briefly discuss the history of Buchweitz's manuscript, present the main result that brings the singularity category of a Gorenstein ring "down to earth," and (if time permits) discuss the significance and impact of Buchweitz's manuscript.

DISCLAIMER

These notes come directly from the manuscript of Buchweitz (1952 - 2017) that was later published into a book with appendices, comments, and errata by Luchezar Avramov, Benjamin Briggs, Srikanth Iyengar, and Janina Letz under the title "Maximal Cohen-Macaulay Modules and Tate Cohomology." We present what Buchweitz himself considered the main result of the manuscript, but in the commutative ring setting.

This is not my own work.

1 History

The first version of the pre-print I’m about to tell you about dates back to in 1986 when Ragnar-Olaf Buchweitz (at U. of Toronto) intended to include the manuscript as part of the conference proceedings of the *Singularities, representation of algebras, and vector bundles* that took place in Lambrecht, 1985. In 2021, the above authors got permission from his spouse to publish this book consisting of his manuscript posthumously. The new authors (listed above) claim this work significantly extends earlier work of Eisenbud on hypersurface singularities and of Tate on group cohomology of finite groups (two things I currently know almost nothing about).

The impact the manuscript had on commutative algebra was slow at first but regained traction in 2004 when Dmitri Orlov rediscovered the stable derived category (the star of the main result) and dubbed it the “singularity category.” In 2008, Buchweitz scanned the unpublished manuscript and put them on the UT website, where they can still be found. In June of 2021, there have been 295 citations attributed to this title, 189 of them since 2015.

As of last night, this book (with appendices from new authors) has been cited 147 times.

2 Math

2.1 Setting the stage

- While Buchweitz works over a not necessarily commutative ring (though noetherian on both sides), let us stick with R is a commutative noetherian ring.
- All complexes will have degree one differentials $X = \cdots \rightarrow X^{i-1} \xrightarrow{\partial^{i-1}} X^i \xrightarrow{\partial^i} X^{i+1} \rightarrow \cdots$ (i.e. cohomological indexing). Here, the sources here share the index of the differential, which is contrary to Buchweitz’s notation, in which the map shares index of the target.

2.2 Categories

1. $\text{Mod}(R)$ is the category of R -modules and R -module homomorphisms $\text{mod}(R)$ for finitely generated R -modules.
2. $\underline{\text{mod}}(R)$ is the “projectively stable” category of finitely generated R -modules, which we obtain from $\text{mod}(R)$ by factoring out all the projective modules from $\text{mod}(R)$. So $\underline{\text{mod}}(R)$ has the same objects as the parent category, but

$$\text{Hom}_{\underline{\text{mod}}(R)}(M, N) := \text{Hom}_R(M, N) / \{f : M \rightarrow N \text{ factors over a projective } R\text{-module}\}.$$

In particular, $M \cong N$ in this category iff there is some f.g. projective R -modules P and Q such that $M \oplus P \cong N \oplus Q$. It is still an additive category.

3. $\text{Ch}(R)$ is the category of (co)chain complexes and chain maps over R .
4. $K(R)$ has the same objects of $\text{Ch}(R)$ but the morphisms are homotopy classes of chain maps.
5. $D_f^b(R)$ is the derived category of R consisting of objects that are cocomplexes over R with bounded and (degree-wise) finitely generated cohomology (over R). We formally invert all quasi-isomorphisms from $\text{Ch}(R)$ so that the a q.i. is an isomorphism in this category. We mark isomorphisms in this category with the q.i symbol $\simeq$. Christensen, Holm, and Foxby put us through ≈ 30 pages of notational hell to localize our morphisms.

- (a) As Zach mentioned, $D_f^b(R)$ is not abelian, but it’s triangulated, which is kind of the next best thing for a homological algebraist. Triangulated categories come with a family of endofunctors $\Sigma^n : D_f^b(R) \rightarrow D_f^b(R)$ called the translation/shift/suspension which, for us, shifts complexes to the left (against the differential) and the sign of the differential inherits a sign of $(-1)^n$. For example, if

M is a complex then $(\Sigma^n M_i) = M_{i-n}$

- (b) Zach talked about the derived category of a dg-ring where the complexes were dg modules, we will not (yet) consider this derived category with extra structure.

Not also my indexing convention is dual to that of Zach's.

2.3 Perfect Complexes

Definition 1. We call a complex of R -modules **perfect** if it is isomorphic in $D_f(R)$ to a bounded complex of finitely generated projective R -modules. (This has nothing to do with the notion of a “perfect module” where $\text{pdim}_R M = \text{grade}_R M =: \min\{t \mid \text{Ext}_R^t(M, R) \neq 0\}$.)

The perfect complexes form an essential (i.e. closed under iso.), full, and triangulated subcategory of $D_f^b(R)$, which we call $\text{Perf}(R)$.

Remark 1. For the experts, perfect complexes are supposed to capture the homological features of complexes induced from regular rings: If S can be represented as a (left) module over a regular ring R (i.e. a ring of finite (right) global dimension) then $\text{Perf}(S)$ is the smallest thick subcategory that contains the image of the functor

$$- \otimes_R^L S : D_f^b(R) \rightarrow D_f^b(S).$$

Fact 1. Let $X \in D^b(R)$. TFAE:

1. X is perfect
2. There is some $i(X) \in \mathbb{Z}$ such that for any $i \geq i(X)$ and any finitely generated R -module M one has

$$\text{Ext}_R^i(X, M) = 0.$$

3. For any exact functor $F : D^b(R) \rightarrow T$ where the target is another triangulated category with $F(R) = 0$, one has $F(X) = 0$.

Of course, facts are things I don't want to prove. If someone asks: For (1) iff (2), one need only notice that X is perfect iff in any of its projective resolutions $F \rightarrow X$, $\text{coker}(\partial_F^i)$ are projective for $i \ll 0$. But F has bounded cohomology, which happens iff $F^{\leq i} \rightarrow \text{coker}(\partial_F^i)$ is contractible for some i , which happens iff (2) holds. For (1) iff (3) see Remark 2.2 and then notice the thick hull of R must contain $\text{Perf}(R)$ which is already thick, and thus we have equality in Remark 2.2 which shows (1) iff (3).

Some remarks are in order:

Remark 2. 1. Here, Ext can be thought of as a Hom in the derived category. In particular, Σ is a triangulated functor, and so

$$\begin{aligned}
\text{Ext}_R^i(X, M) &:= H^i \mathbf{R}\text{Hom}_R(X, M) \\
&\cong H^0 \Sigma^i \mathbf{R}\text{Hom}_R(X, M) \\
&\cong H^0 \mathbf{R}\text{Hom}_R(X, \Sigma^i M) \\
&\cong H^0 \text{Hom}_R(P(X), P(\Sigma^i M)) \\
&\cong \text{Hom}_{K(R)}(P(X), P(\Sigma^i M)) && \text{(The above is the Hom ex.)} \\
&\cong \text{Hom}_{\mathbf{D}^b(R)}(X, \Sigma^i M). && \text{(This is why we needed to use } \Sigma^i M \simeq P(\Sigma^i M))
\end{aligned}$$

Recall that the 0-cycles in the Hom complex are just chain maps, and the 0-boundaries of the Hom complex are nullhomotopies, so $H^0 \text{Hom}_R(-, -) \cong \text{Hom}_{K(R)}(-, -)$. Finally, the last isomorphism comes from the (unproven) fact that the map $\text{Hom}_{K(R)}(P(X), P(M)) \rightarrow \text{Hom}_{\mathbf{D}^b(R)}(X, M)$ given by a mapping of fractions is actually an isomorphism.

2. Condition (3) characterizes the smallest thick subcategory of $\mathbf{D}_f^b(R)$ (this is a kind of universal property) that contains R , and so the equivalence of (1) and (3) is meant to establish $\text{Perf}(R) = \text{Thick}(R)$.

The main object of study is as follows:

Definition 2 (Buchweitz, 1986 and Orlov, 2004). *To a noetherian ring R , Buchweitz calls the Verdier quotient category*

$$\underline{D}_f^b(R) := D_f^b(R) / \text{Perf}(R) \quad (D_{\text{sg}}(R) \text{ in Orlov's notation.})$$

*the **stabilized derived category of R** . To an algebraic variety X , Orlov calls the quotient of the derived category of coherent sheaves on X by the perfect complex subcategory*

$$D_{\text{sg}}(X) = D_f^b(\text{coh}(X)) / \text{Perf}(X)$$

*the **singularity category of X** . Even for a ring, we will use Orlov's more modern name and notation for the first Verdier quotient.*

Remark 3. *Buchweitz calls this stabilized because for $i \gg 0$ (with the Gorenstein hypothesis, as the others point out), one has $\text{Hom}_{\underline{D}_f^b(R)}(X, \Sigma^i Y) = \text{Hom}_{D_f^b(R)}(X, \Sigma^i Y) = \text{Ext}_R^i(X, Y)$ for all $i \gg 0$.*

Orlov calls this the singularity category because when X is a smooth variety, $D^b(\text{coh}(X)) = \text{Perf}(X)$. And thus, when X is singular this verdier quotient is not trivial.

2.4 MCMs

Because Buchweitz is working in the noncommutative and often not local setting, he has a different (weaker) notion of MCM which may be of use to us. In our commutative local case, we claim this notion is the same.

Proposition 1. *Suppose $(R, \mathfrak{m}, k)$ is a Gorenstein local ring and M is a finitely generated R -modules. Then the following are equivalent:*

1. *M is MCM*
2. *$\text{Ext}_R^i(M, R) = 0$ for all $i \neq 0$.*

Remark 4. When R is merely a ring noetherian on both sides with finite injective dimension (on both sides) then Buchweitz calls (2) the definition of M being MCM. It is a Theorem of A. Zaks that if the injdim of R is finite on both sides, then it is the same.

Proof. (1) $\implies$ (2) follows from Theorem 6.28 in Eloisa's 906 notes from last semester (by just adding the fact that R is Gorenstein so $\omega = R$ is the canonical module). This theorem is a more general result concerning the (non)vanishing of $\text{Ext}_R^i(M, \omega)$, where M is CM and the nonvanishing index is $i = \dim(R) - \dim(M)$. The proof is 2 pages long; it is one of the results we probably spent close to a whole class on, or that she skipped entirely because it was too long. The proof is by induction on $\dim(M)$ (of finite dimension since M f.g. tells us dimension can be computed as the maximal length of saturated chains of primes in the support of M , hence $\dim R \geq \dim M$). In the base case, you also induce on the length of M and either use that $M \cong k$ or the SES $k \rightarrow M \rightarrow N$ (where N is the cokernel) inducing the LES on $\text{Ext}_R^*(-, \omega)$. In the (big) inductive step, you kill an M -regular element to knock depth and dimension down and apply the same functor to the SES $M \xrightarrow{x} M \rightarrow M/(x)$ and examine the induced LES on Ext .

Local duality actually allows us to give an easier proof (that is, if you consider local duality easier): Define the Matlis functor $(-)^{\vee} := \text{Hom}_R(-, E_R(k))$. If $d = \dim R$, and $(R, \mathfrak{m}, k)$ admits a canonical module ω , then local duality gives us $H_{\mathfrak{m}}^i(M) \cong \text{Ext}_R^{d-i}(M, \omega)^{\vee}$ for any f.g. R -module M , and hence

$$\begin{aligned} \text{Ext}_R^{d-i}(M, \omega) = 0 \quad \forall i \neq d &\iff \text{Ext}_R^{d-i}(M, \omega)^{\vee} = 0 \quad \forall i \neq d \quad (\text{Matlis functor is faithful}) \\ &\iff H_{\mathfrak{m}}^i(M) = 0 \quad \forall i \neq d. \quad (\text{Thm 4.27 in Jack's LC notes}) \\ &\iff \text{depth}(M) = \dim(M) = d = \dim(R) \\ &\iff M \text{ is MCM} \end{aligned}$$

because LC detects depth/dimension. □

Definition 3. The (f.g.) maximal Cohen-Macaulay modules in $\text{mod}(R)$ form a full subcat-

egory that we denote $\text{MCM}(R)$. The image of $\text{MCM}(R)$ in $\underline{\text{mod}}(R)$ is denoted $\underline{\text{MCM}}(R)$.

2.5 Complete resolutions

Definition 4. *M an R-module. A **complete resolution** of M over R is an exact complex of finitely generated projective R-modules $A = \cdots \rightarrow A^i \xrightarrow{d^i} A^{i+1} \rightarrow \cdots$ such that $\text{coker}(d_A^{-1} : A^{-1} \rightarrow A^0) = M$. We call the truncations $A^{\leq 0}$ the associated projective resolution of M and $A^{\geq 1}$ the associated co-projective resolution of M.*

Remark 5. *Note M is necessarily finitely generated if it admits a complete resolution (R is noetherian after all) and a finitely generated module admits a complete resolution if and only if M admits a projective co-resolution.*

Example 1. *Let $(R, \mathfrak{m}, k)$ be artinian Gorenstein, and M a f.g. R-module. Let $F \rightarrow M$ be the minimal free resolution of M and $M \rightarrow E$ the minimal injective resolution of M. Since $\text{Spec}(R) = \{\mathfrak{m}\}$, $E^i \cong E_R(k)^{\mu_i(\mathfrak{m}, M)}$. In fact since R is injective in this case, one can show that $R \cong E_R(k)$ [a 906 exercise]. Recall that $\mu_i(\mathfrak{m}, R) = \dim_k \text{Ext}_R^i(k, R)$ is finite whenever M is finitely generated, and so E is a complex of f.g. frees. Then reindexing and connecting $F \rightarrow E$ gives us a complete resolution of M :*

$$\cdots \rightarrow F_{-1} \xrightarrow{\partial_F^{-1}} F_0 \xrightarrow{0} E_0 \xrightarrow{\partial_E^0} \cdots$$

This is an exact complex of f.g. frees and $\text{coker}(\partial_F^{-1}) = M$ by construction.

Definition 5. *We denote by $\underline{\text{APC}}(R)$ the homotopy category of (unbounded) exact complexes of finitely generated projective R-modules. Note these are necessarily complete resolutions of the appropriate “syzygy.”*

For any complex of R-modules X, we define the **i**th syzygy functor $\Omega_i(-)$ defined by $\Omega_i(X) = \text{coker}(\partial_X^{-i-1} : X^{-i-1} \rightarrow X^{-i})$ for any $i \in \mathbb{Z}$. Note that if $f : X \rightarrow Y$ is a morphism of complexes of f.g. projective R-modules which is nullhomotopic, then for any i the induced

morphism of R -modules $\Omega_i(f) : \Omega_i(X) \rightarrow \Omega_i(Y)$ factors over a finitely generated projective R -module. In particular, this shows that:

Lemma 1. *Each syzygy functor induces a functor (with the same name) $\Omega_i : \underline{\text{APC}}(R) \rightarrow \underline{\text{mod}}(R)$. [In fact, as part of the proof of the main result, one must show that we can restrict the target category to be $\underline{\text{MCM}}(R)$.]*

2.6 The main result

Note that the composition of functors $\underline{\text{mod}}(R) \rightarrow D_f^b(R) \rightarrow D_{\text{sg}}(R)$ factors uniquely over the projection functor $\underline{\text{mod}}(R)$, and so one obtains a canonical inclusion functor $\iota_R : \underline{\text{mod}}(R) \rightarrow D_{\text{sg}}(R)$.

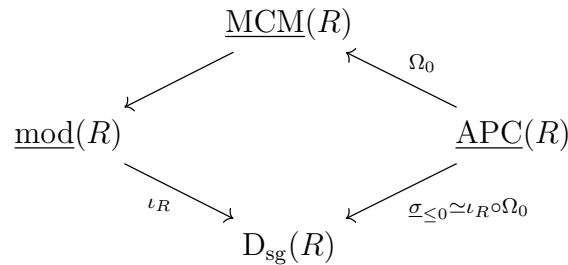
We present the main result:

Theorem 1. *Let R be a Gorenstein ring (or R left and right noetherian with finite injdim on either side). Then*

1. *The syzygy functor $\Omega_0 : \underline{\text{APC}}(R) \rightarrow \underline{\text{MCM}}(R)$ induces an equivalence of categories.*
2. *The restriction of the inclusion functor $\iota_R : \underline{\text{MCM}}(R) \rightarrow D_{\text{sg}}(R)$ induces an equivalence of categories.*

We only give the proof strategy.

Proof Strategy. The first step is to argue the image of Ω_0 in fact lies in $\underline{\text{MCM}}(R)$. Then we have a diagram of categories and functors below.



One argues that Ω_0 is essentially surjective (surjective on objects) [this is where the Gorenstein hypothesis is used, it implies that MCM's are reflexive, and this is used to construct the complete resolution that maps to your MCM] and a full functor (surjective on morphisms). Buchweitz mentions that one could finish this by showing Ω_0 is also faithful (injective on morphisms), but he would rather show:

There is a directed system of functors $\underline{\text{APC}}(R) \rightarrow \text{D}_{\text{sg}}(R)$ formed by the naive truncations $\cdots \xrightarrow{\simeq} \underline{\sigma}_{\leq k} \xrightarrow{\simeq} \underline{\sigma}_{\leq k-1} \xrightarrow{\simeq} \cdots$ whose transition morphisms are all isomorphisms. Hence, the inverse limit exists and $\underline{\sigma}_{\leq} := \varprojlim_k \underline{\sigma}_{\leq k}$ is an exact functor of triangulated categories. Buchweitz shows this limit is in fact an equivalence of categories. He previously argued that there are natural isomorphisms of functors

$$\underline{\sigma}_{\leq} \xrightarrow{\simeq} \underline{\sigma}_{\leq 0} \xrightarrow{\simeq} \iota_R \circ \Omega_0.$$

So this equivalence would show that Ω_0 is faithful, and thus an equivalence as claimed. But then ι_R would also have to be an equivalence of categories, which would show the second claim. $\square$

2.7 Broader Impacts

The authors in appendix C give some of the developments in the 35ish years since Buchweitz's manuscript.

- There are some connections with the BGG (Bernstein-Gelfand²) correspondence. [relates the graded MCM category over the exterior algebra $\Lambda = \bigwedge(y_1, \dots, y_n)$ to the derived category of projectives over $S = k[x_1, \dots, x_n]$, which works because S and Λ are Koszul dual to each other].
- Orlov's work in which he rediscovered the singularity category is related to homological mirror symmetry (a physics notion).

- There are connections with matrix factorizations.
- There are connections with the theory of Gorenstein-projectives.
- Stable cohomology and stable Ext, stable categories.
- dg-categories: $D_{\text{sg}}(R)$ is the homotopy category of a pre-triangulated dg-category (a category enriched over dg-modules) meaning the hom-sets in $D_{\text{sg}}(R)$ can be upgraded to be viewed as dg-modules. This allows one to have access to more invariants to an algebra R than would otherwise be available.